








Statistics of Financial Markets

Seminar 14: Crypto assets



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); Abnormal Return
- **CI**: Confidence Interval
- **COVID-19**: COronaVIrus Disease 2019
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund
- **FTX**: FTX Trading Ltd. (crypto exchange, bankrupt in November 2022)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **HS**: Historical Simulation
- **LR**: Likelihood Ratio
- **MDD**: Maximum DrawDown
- **SVB**: Silicon Valley Bank
- **US**: United States
- **USDC**: USD Coin
- **UST**: TerraUSD

Today's Question and Seminar Structure

- Question of the seminar
  - **how risky is a crypto position, and how stable is a "stable" coin?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the Exercise Map (next slide)
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	annualising a crypto series with 365 and with 252 days	Solved, Proposed	A1
A3, A4	the maximum drawdown of a price path and the gain needed to recover	Solved, Proposed	A3
A5, A6	peg deviations of USDC and DAI in March 2023, in basis points	Solved, Proposed	A5
A7, A8	VaR 1% and ES 2.5% of a Bitcoin and of an Ethereum position	Solved, Proposed	A7
B1, B2	moments, annualisation and tail index: Bitcoin and S&P 500; Ethereum and Solana	Solved, Proposed	B1
B3, B4	correlation in calm and crisis periods: Bitcoin and S&P 500; Bitcoin and gold, Ethereum and S&P 500	Solved, Proposed	B3
B5, B6	the peg of USDC; USDT and DAI	Solved, Proposed	B5
B7, B8	VaR, ES and a backtest: Bitcoin; Ethereum	Solved, Proposed	B7
C1, C2	did the spot ETFs change Bitcoin? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model in the last column
- Part A: calculations on paper; Part B: real data, in the notebook, each ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data Used

Series	Source	Price used	Period
Bitcoin, Ethereum, Solana	EODHD	close, 7 days a week	2014/2016/2020–2026
USDT, USDC, DAI	EODHD	close and daily low	2021–2026
S&P 500, gold (XAU/USD)	EODHD	close, weekdays	2014–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$, each series on its own calendar; the last day is 18 September 2026
- Two series together: join the prices on common days first, then compute returns
- In the notebook: `returns('btc')`, `joint_returns('btc', 'sp500')`, `describe(k)`, `peg_stats('usdc')`; no account or key is needed

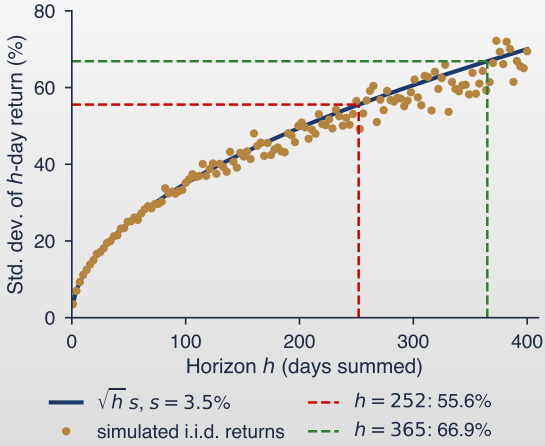
Prerequisites for Today (1/16): Crypto Assets and Annualisation

- **Crypto asset**: a digital asset whose ownership is recorded on a **blockchain**, a public chain of blocks of transactions
  - Bitcoin (2009): fixed supply of 21 million coins; Ethereum (2015): runs programs (smart contracts)
  - traded on **exchanges** (trading platforms) 24 hours a day, 7 days a week
- **Annualisation** with A observations a year (i.i.d. returns) and the **Sharpe ratio** (risk-free rate 0):

$$\mu_{an} = A\,\bar{r}, \qquad \sigma_{an} = \sqrt{A}\,s, \qquad \mathrm{SR} = \frac{\mu_{an}}{\sigma_{an}} = \sqrt{A}\,\frac{\bar{r}}{s}$$

- Notation
  - $\bar{r}$: the mean daily return; s: the standard deviation of daily returns; i.i.d.: independent and identically distributed
  - crypto: $A = 365$, $\sqrt{365} = 19.105$; equities: $A = 252$, $\sqrt{252} = 15.875$
  - the mean and the volatility must use the same A, otherwise the Sharpe ratio is wrong

Prerequisites for Today (2/16): The Square-Root Rule



- For i.i.d. daily returns, the variance of a sum of h days is h times the daily variance:
  - $\mathrm{Var}(r_1 + \cdots + r_h) = h\,s^2 \;\Rightarrow\; \mathrm{sd} = \sqrt{h}\,s$
- Chart: 60 simulated years with $s = 3.5\%$
  - dots: the standard deviation of simulated h-day returns; line: $\sqrt{h}\,s$
  - a crypto year is $h = 365$ trading days, not 252
- Reading
  - with 252 the annual volatility is too low by the factor $\sqrt{252/365} = 0.83$, whatever s is
  - the rule needs independence; with volatility clustering it is an approximation

Prerequisites for Today (3/16): Moments of a Sample

- For n daily returns $r_1, \ldots, r_n$ (in %): the sample mean and standard deviation

$$\bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad s = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2}$$

- **Skewness** and **excess kurtosis** of the standardised returns $z_t = (r_t - \bar{r})/s$:

$$\hat{S} = \frac{1}{n}\sum_{t=1}^{n} z_t^3, \qquad \hat{K} = \frac{1}{n}\sum_{t=1}^{n} z_t^4 - 3$$

- Reading
  - $\hat{S} < 0$: large falls are more frequent or larger than large rises
  - $\hat{K} = 0$ for the Normal distribution; $\hat{K} > 0$: **heavy tails**, more extreme days than under the Normal distribution with the same variance
  - both depend on a few extreme days, so they are unstable across samples

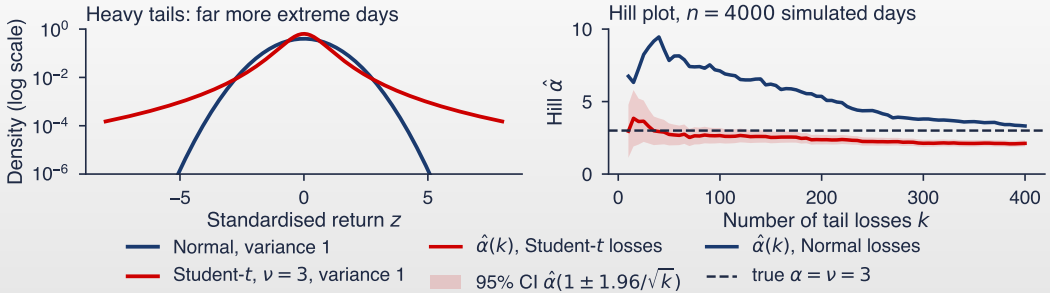
Prerequisites for Today (4/16): The Hill Tail Index

- **Power tail**: for large losses, $P(\text{loss} > x) \approx c\, x^{-\alpha}$
  - α: the **tail index**; $c > 0$: a constant; a smaller α means a heavier tail
  - moments of order $p < \alpha$ exist; e.g. $\alpha < 4$: the kurtosis is not finite
- **Hill estimator** (Chapter 5), from the k largest losses $X_{(1)} \geq X_{(2)} \geq \ldots$:

$$\hat{\alpha} = \Big[\frac{1}{k}\sum_{i=1}^{k} \ln \frac{X_{(i)}}{X_{(k+1)}}\Big]^{-1}, \qquad 95\%\ \text{CI}: \ \hat{\alpha}\Big(1 \pm \frac{1.96}{\sqrt{k}}\Big)$$

- Notation and reading
  - losses: $X_t = -r_t$ for the left tail, $X_t = r_t$ for the right tail; $X_{(k+1)}$: the threshold of the tail
  - the bracket is the mean log distance of the k largest losses from the threshold: far apart means a heavy tail
  - here $k = 2.5\%$ of n; two intervals that overlap do not separate the two tails

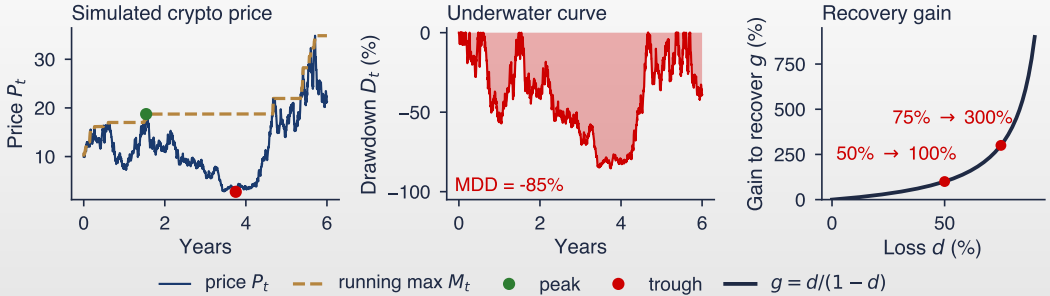
Prerequisites for Today (5/16): Heavy Tails and the Hill Plot



- Left: on a log scale the Student-t density with $\nu = 3$ lies far above the Normal one beyond 4 standard deviations, with the same variance
- Right: **Hill plot**, $\hat{\alpha}(k)$ against k, on 4000 simulated days
  - Student-t losses: a flat region near the true $\alpha = 3$; Normal losses: $\hat{\alpha}$ keeps changing (no power tail)

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Prerequisites for Today (6/16): Drawdown and the Gain to Recover



- **Drawdown**: $D_t = P_t/M_t - 1 \leq 0$, with the running maximum $M_t = \max_{s \leq t} P_s$ (the highest price up to t); **maximum drawdown** $\text{MDD} = \min_t D_t$
- **Gain to recover** a loss $d = |D_t|$: $(1-d)(1+g) = 1$, so $g = d/(1-d)$
  - right: g grows without bound as $d \to 1$; a fall of 50% needs +100%, a fall of 75% needs +300%

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Prerequisites for Today (7/16): Correlation and the Fisher z Test

- **Sample correlation** of two return series on n common days (prices joined first, then log returns):

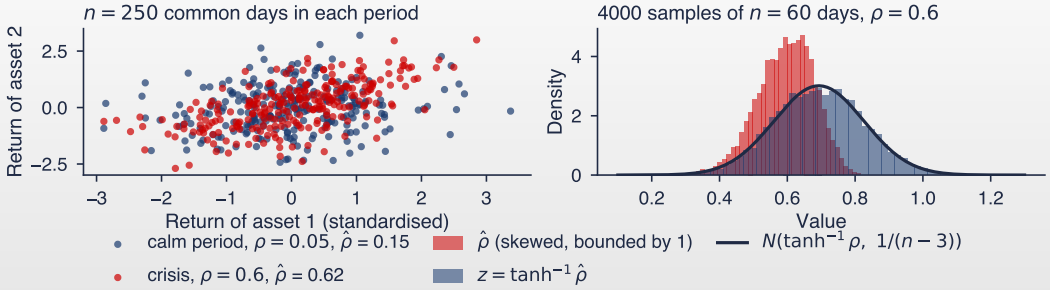
$$\hat{\rho} = \frac{\sum_t (x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_t (x_t - \bar{x})^2 \; \sum_t (y_t - \bar{y})^2}} \in [-1, 1]$$

- **Fisher** z transform and the test of equal correlation in two independent periods:

$$z = \tanh^{-1} \hat{\rho} = \tfrac{1}{2} \ln \frac{1 + \hat{\rho}}{1 - \hat{\rho}}, \qquad Z = \frac{z_2 - z_1}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}$$

- Reference distributions and reading
  - $z \approx N(\tanh^{-1} \rho,\ 1/(n-3))$; 95% CI for ρ: $\tanh(z \pm 1.96/\sqrt{n-3})$
  - under $\rho_1 = \rho_2$, $Z \approx N(0,1)$; $|Z| > 1.96$: the two correlations differ at 5%
- Other quantities in B3, B4
  - weekly return: $100 \ln(P_{\text{Fri}}/P_{\text{previous Fri}})$; cumulative log return of an episode: $\sum_t r_t$ over its days
  - **safe haven** Baur & Lucey (2010): an asset that does not fall (or rises) when the stock market crashes; check its mean return on the worst 5% of equity days

Prerequisites for Today (8/16): Correlation in Calm and Crisis Periods



- Left: 250 simulated days of two assets in a calm period ($\rho = 0.05$) and in a crisis ($\rho = 0.6$): the cloud stretches along the diagonal
- Right: over 4000 samples of 60 days, $\hat{\rho}$ is skewed (it cannot exceed 1), while $z = \tanh^{-1}\hat{\rho}$ follows the Normal curve: this is why the test uses z

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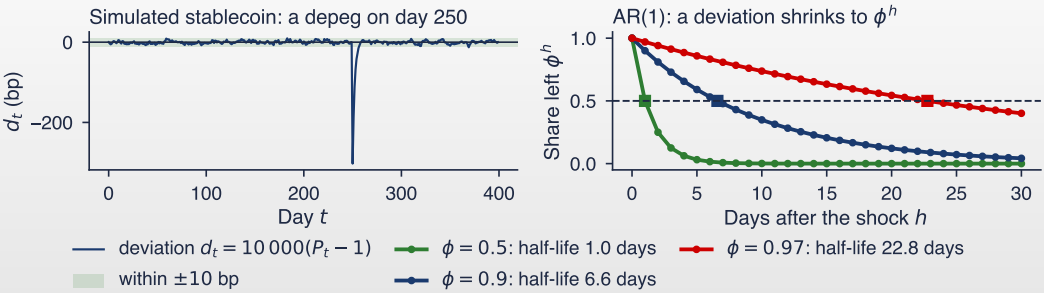
Prerequisites for Today (9/16): Stablecoins and Basis Points

- **Stablecoin**: a crypto asset designed to trade at a fixed price, usually 1 USD (the **peg**)
  - fiat-backed (USDT, USDC: dollar reserves), crypto-backed (DAI: crypto collateral), algorithmic (UST: no reserves, collapsed in May 2022)
  - **depeg**: the price moves away from the peg; March 2023: Silicon Valley Bank (SVB), which held part of the USDC reserves, failed
- **Basis point** (bp) $= 0.01\% = 0.0001$; **peg deviation** of the price P_t (USD) on day t:

$$d_t = 10\,000\,(P_t - 1) \text{ bp}$$

- Reading
  - $d_t < 0$: below the peg; $P_t = 0.9970$ gives $d_t = -30$ bp, i.e. -0.30%
  - a holder who sells Q coins at price $P < 1$ loses $Q(1 - P)$ dollars
  - summary of d_t: its standard deviation, the median of $|d_t|$ and the shares of days beyond 10 and 50 bp

Prerequisites for Today (10/16): Persistence of a Deviation



- **AR(1)** model: $d_t = c + \phi\, d_{t-1} + e_t$, $|\phi| < 1$; c: a constant; ϕ: the share of yesterday's deviation still present today; e_t: a new shock with mean 0
- **Half-life** $h = \ln 0.5 / \ln \phi$ days: after h days a deviation has shrunk to $\phi^h = 0.5$ of its size (squares, right)
  - left: a simulated coin with $\phi = 0.6$ falls 300 bp on day 250 and is back at the peg within a few days

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Prerequisites for Today (11/16): VaR and ES

- Value-at-Risk and Expected Shortfall at level α (VaR 1%, ES 2.5%), as positive losses:

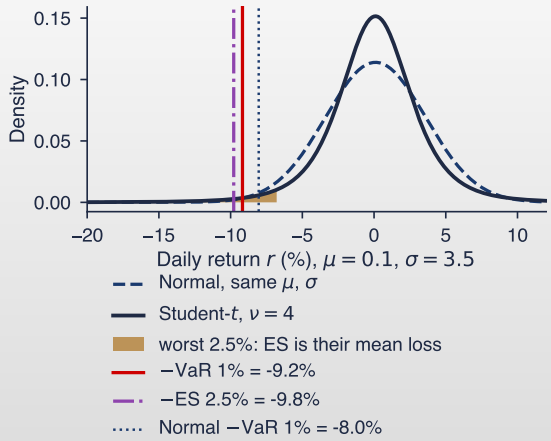
$$\mathrm{VaR}_\alpha = -q_\alpha, \qquad \mathrm{ES}_\alpha = -E[r \mid r \le q_\alpha]$$

- Notation
  - q_α: the α-quantile of the return r, $P(r \le q_\alpha) = \alpha$; ES: the mean loss on the days beyond the quantile
- Normal returns with mean μ and standard deviation σ:

$$\mathrm{VaR}_{\mathbf{1\%}} = -\mu + 2.326\,\sigma, \qquad \mathrm{ES}_{\mathbf{2.5\%}} = -\mu + \sigma\,\frac{\varphi(1.960)}{0.025}$$

- Notation
  - 2.326, 1.960: the 99% and 97.5% quantiles of the standard Normal distribution; φ: its density, $\varphi(1.960) = 0.0584$
- In money and over h days
  - position W: simple returns $W \times \mathrm{VaR}/100$; log returns $W(1 - e^{-\mathrm{VaR}/\mathbf{100}})$
  - h days (i.i.d., mean 0): $\sqrt{h}\,\mathrm{VaR}$; crypto: h calendar days

Prerequisites for Today (12/16): VaR and ES with Heavy Tails

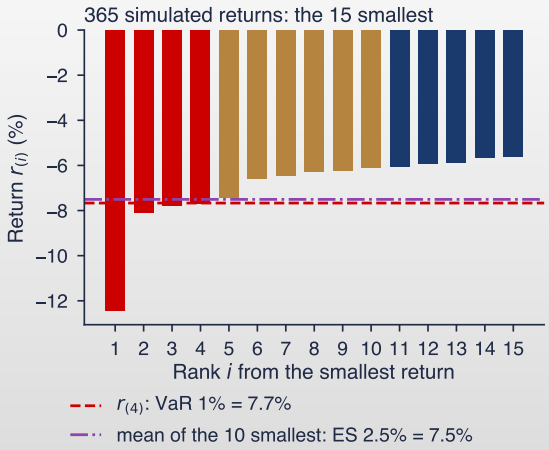


- **Student-t** returns, fitted by maximum likelihood (location μ, scale s, degrees of freedom ν):

$$\mathrm{VaR}_\alpha = -(\mu + s\, t_\nu^{-1}(\alpha))$$

- Notation
  - $t_\nu^{-1}(\alpha)$: the α-quantile of the standard Student-t; variance $s^2\nu/(\nu-2)$
  - ES: $-\mu + s\,\frac{f_\nu(t_\alpha)}{\alpha}\,\frac{\nu+t_\alpha^2}{\nu-1}$, $t_\alpha = t_\nu^{-1}(\alpha)$, f_ν: the t density
- Chart: same mean and variance
  - the Student-t VaR 1% (red) lies beyond the Normal one (dotted)
  - ES 2.5% is the mean loss in the shaded 2.5% tail

Prerequisites for Today (13/16): Historical Simulation

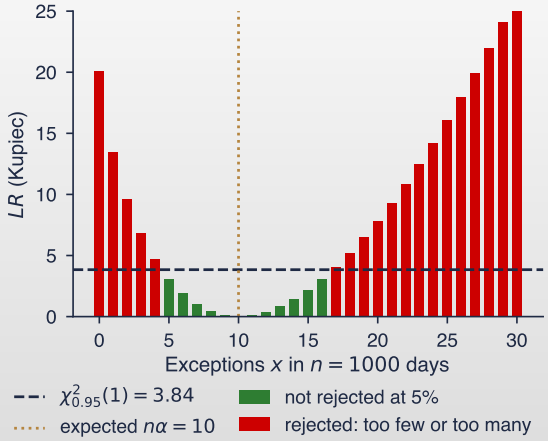


- **Historical simulation** (HS) on the last n returns, sorted: $r_{(1)} \leq r_{(2)} \leq \dots$
  - no distribution is assumed: the past days are the scenarios
- VaR and ES at level α:

$$k = \lceil n\alpha \rceil, \quad \mathrm{VaR} = -r_{(k)}, \quad \mathrm{ES} = -\frac{1}{k}\sum_{i=1}^{k} r_{(i)}$$

- Notation and reading
  - $\lceil \cdot \rceil$: round up; $r_{(k)}$: the k-th smallest return
  - chart: $n = 365$; VaR 1%: $k = 4$ (red bars); ES 2.5%: the mean of $k = 10$ (red and ochre)
  - ES and VaR at different levels need not be ordered

Prerequisites for Today (14/16): Backtesting: the Kupiec Test

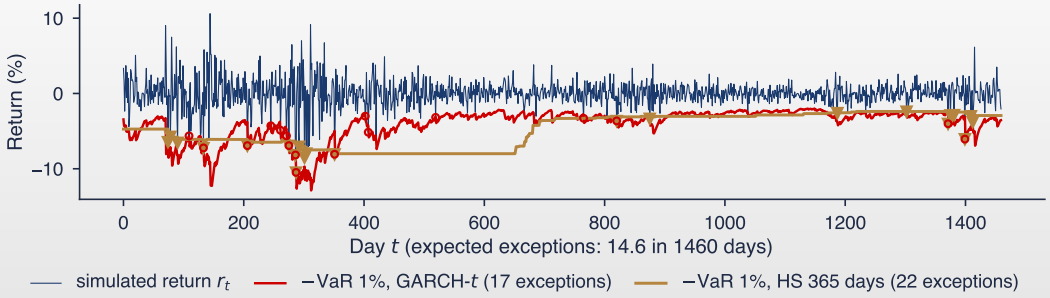


- **Exception**: a day with $r_t < -\mathrm{VaR}_t$; x exceptions in n days Kupiec (1995)
  - H_0: each day is an exception with probability $\alpha = 0.01$
- Likelihood ratio:

$$LR = -2\big[\ln L(\alpha) - \ln L(x/n)\big], \quad L(p) = p^x(1-p)^{n-x}$$

- Notation and reading
  - $L(p)$: the likelihood of x exceptions if each day has probability p; x/n: the observed rate
  - $LR \sim \chi^2(1)$ under H_0; reject at 5% if $LR > 3.84$
  - chart, $n = 1000$: 5 to 16 exceptions pass (green); too few or too many fail

Prerequisites for Today (15/16): GARCH(1,1)-t VaR



- GARCH(1,1)-t (Chapter 9): $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; σ_t^2: the conditional variance; $\varepsilon_{t-1} = r_{t-1} - \mu$: yesterday's shock; $\omega, \alpha, \beta \geq 0$; persistence $\alpha + \beta$
- VaR for tomorrow $= -(\mu + \sigma_{t+1} q_{1\%})$; $q_{1\%}$: the 1% quantile of the Student-t rescaled to variance 1
  - simulated GARCH returns: the GARCH VaR (red) moves at once with volatility; the HS VaR on 365 days (ochre) moves in steps and its exceptions cluster

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Prerequisites for Today (16/16): Weekends and the Brown–Forsythe Test

- **Weekend effect** in the variance (C1): crypto trades on Saturday and Sunday, equities do not
  - weekend/weekday variance $= s^2_{\textbf{weekend}} / s^2_{\textbf{weekday}}$; below 1: calmer weekends
  - weekend share of squared returns $= \sum_{t \in \textbf{weekend}} r_t^2 / \sum_t r_t^2$; $2/7 = 28.6\%$ if every day were alike
- **Brown–Forsythe** test of equal variance in g groups, on the distances from the group medians $Z_{ij} = |r_{ij} - \tilde{r}_i|$:

$$W = \frac{N-g}{g-1} \cdot \frac{\sum_i n_i (\bar{Z}_i - \bar{Z})^2}{\sum_i \sum_j (Z_{ij} - \bar{Z}_i)^2} \sim F(g-1,\ N-g)$$

- Notation and reading
  - r_{ij}: return j of group i (e.g. before and after a date); $\tilde{r}_i$: its median; n_i: its size; $N = \sum_i n_i$
  - $\bar{Z}_i$, $\bar{Z}$: the group and overall means of Z; medians make the test robust to heavy tails; a small p-value: the variances differ

A1: Annualising Bitcoin [Solved]

Task

- Bitcoin daily log returns: mean $\bar{r} = 0.12\%$, standard deviation $s = 3.5\%$. An analyst uses the equity convention of 252 days.
- 1. Compute the annual mean and volatility with 365 days.
- 2. Compute them with 252 days and say by how much they are understated.
- 3. Compute the Sharpe ratio (risk-free rate 0) correctly and with the mean $\times 252$ and the volatility $\times \sqrt{365}$.
- Report: six numbers and one sentence.

Solution

- 1. $365 \times 0.12 = 43.8\%$; $\sqrt{365} \times 3.5 = 19.10 \times 3.5 = 66.9\%$
- 2. $252 \times 0.12 = 30.2\%$ (31.0% too low); $\sqrt{252} \times 3.5 = 55.6\%$ (16.9% too low)
- 3. Correct: $43.8/66.9 = 0.66$; mixed: $30.2/66.9 = 0.45$
- The 252-day convention makes Bitcoin look calmer; mixing conventions makes it look worse.

Back Exercise list

A2: Annualising Ethereum [Proposed]

Solution discussed in class

Task

- Ethereum daily log returns: $\bar{r} = 0.20\%$, $s = 5.0\%$. Model: A1.
- 1. Compute the annual mean and volatility with 365 and with 252 days.
- 2. Compute the ratio of the two volatilities.
- 3. Compute the correct and the mixed Sharpe ratio.
- Report: seven numbers and one sentence.

Back Exercise list

A3: The Maximum Drawdown of a Price Path [Solved]

Task

- Monthly closes of a crypto asset, in thousand USD: 20, 35, 69, 50, 30, 16, 25, 45, 73.
- 1. Write the running maximum M_t and the drawdown D_t of each month.
- 2. Give the maximum drawdown and the gain needed to recover from the trough.
- 3. Say in which month the asset recovers.
- Report: two numbers, a list and one sentence.

Solution

- 1. M_t: 20; 35; 69; 69; 69; 69; 69; 73; D_t (%): 0.0; 0.0; 0.0; -27.5; -56.5; -76.8; -63.8; -34.8; 0.0
- 2. Maximum drawdown $16/69 - 1 = -76.8\%$; gain needed $69/16 - 1 = 331.25\%$
- 3. Month 9 ($73 > 69$): a new record high.
- A fall of about three quarters needs more than a quadrupling to recover.

Back Exercise list

A4: A Second Price Path [Proposed]

Solution discussed in class

Task

- Quarterly closes of Ethereum, in thousand USD: 1.4, 4.8, 3.2, 0.9, 1.2, 2.5, 4.9. Model: A3.
- 1. Write the running maximum and the drawdowns.
- 2. Give the maximum drawdown and the gain needed to recover.
- 3. Say in which quarter the asset recovers.
- Report: two numbers, a list and one sentence.

Back Exercise list

A5: USDC in March 2023 [Solved]

Task

- USDC daily closes: 10 March 2023 0.9995, 11 March 0.9715, 12 March 0.9921, 13 March 0.9989; lowest daily low 0.8774 (11 March).
- 1. Compute the peg deviation of each close in basis points.
- 2. Compute the deviation of the lowest low.
- 3. Compute the loss of a holder of 1 000 000 USDC who sold at the worst close, and at the lowest low.
- Report: five deviations, two losses and one sentence.

Solution

- 1. $10\,000(P-1)$: -5, -285, -79, -11 bp
- 2. $10\,000 \times (0.8774 - 1) = -1226$ bp
- 3. $1\,000\,000 \times (1 - 0.9715) = 28\,500$ USD; at the low: 122 600 USD
- Who sold in panic on Saturday lost; who held until Monday, after the deposits were guaranteed, lost almost nothing.

Back Exercise list

A6: DAI in March 2023 [Proposed]

Solution discussed in class

Task

- DAI daily closes: 10 March 2023 0.9990, 11 March 0.9739, 12 March 0.9926, 13 March 0.9989; lowest daily low 0.8970. Model: A5.
- 1. Compute the four closing deviations and the deviation of the low in bp.
- 2. Compute the loss on 1 000 000 DAI sold at the worst close and at the low.
- 3. Explain why DAI, a crypto-backed coin, followed USDC.
- Report: five deviations, two losses and one sentence.

Back Exercise list

A7: VaR of a Bitcoin Position [Solved]

Task

- A position of 100 000 USD in Bitcoin.
- 1. With Normal daily simple returns ($\mu = 0.10\%$, $\sigma = 3.5\%$), compute VaR 1%, ES 2.5% and the 10-day VaR 1% in % and in USD.
- 2. The 10 smallest of the last 365 log returns (19 September 2025 – 18 September 2026), in %: -15.23; -7.23; -6.77; -6.69; -5.46; -5.43; -5.32; -4.76; -4.69; -4.62. Compute the historical VaR 1% and ES 2.5% in % and in USD.
- Report: ten numbers and one sentence.

Solution

- 1. VaR $= -0.10 + 2.326 \times 3.5 = 8.04\%$ (8 042 USD); ES $= -0.10 + 3.5 \times 0.0584/0.025 = 8.08\%$ (8 082 USD); 10 days: $\sqrt{10} \times 2.326 \times 3.5 = 25.75\%$ (25 748 USD)
- 2. $k = \lceil 3.65 \rceil = 4$: VaR 1% $= 6.69\%$, $100\,000(1 - 0.9353) = 6\,471$ USD; $k = \lceil 9.125 \rceil = 10$: ES 2.5% $= 66.20/10 = 6.62\%$ (6 406 USD)
- ES 2.5% can be below VaR 1%: ES $\geq$ VaR holds only at the same level.

Back Exercise list

A8: VaR of an Ethereum Position [Proposed]

Solution discussed in class

Task

- A position of 100 000 USD in Ethereum. Model: A7.
- 1. With Normal simple returns ($\mu = 0.20\%$, $\sigma = 5.0\%$), compute VaR 1%, ES 2.5% and the 10-day VaR 1%, in % and in USD.
- 2. The 10 smallest of the last 365 log returns, in %: -16.27; -12.83; -11.28; -10.00; -8.99; -8.22; -8.20; -7.54; -7.52; -7.11. Compute the historical VaR 1% and ES 2.5% in % and in USD.
- Report: ten numbers and one sentence.

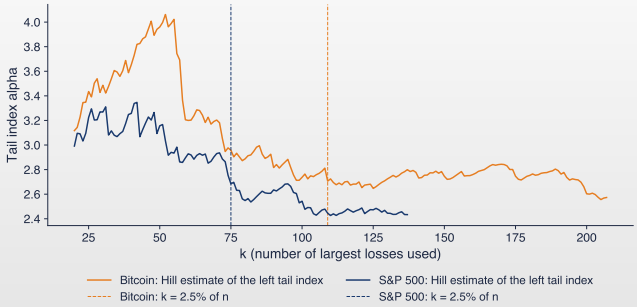
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B1: Bitcoin Against the S&P 500 [Solved]

- **Question**: is Bitcoin's distribution different from that of the S&P 500 only in scale, or also in the shape of its tails?
- **Inputs**: Bitcoin and S&P 500 closes since September 2014, daily log returns in %, each on its own calendar
- **Tasks**
  1. Compute the mean, the standard deviation, the skewness and the excess kurtosis of each series.
  2. Annualise the mean and the volatility with the actual frequency and with 252 days.
  3. Compute the Hill index of the left and of the right tail with $k = 2.5\%$ of n and its 95% interval.
  4. Draw the Hill plot of the losses for both series.
  5. Interpretation: are Bitcoin's tails heavier than those of the S&P 500?
- **Report**: a table of 12 numbers, the chart and two sentences
- Notebook: section B1

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B1: Solution (1/2) [Solved]



- Means: 0.118% and 0.044% a day; annual means 43.1% ($A = 365.3$) and 11.2% ($A = 251.4$)

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B1: Solution (2/2) [Solved]

	n	sd	skew.	exc. kurt.	vol. (actual A)	vol. (252)	$\hat{\alpha}$ left [95% CI]	$\hat{\alpha}$ right
Bitcoin	4 384	3.49	−0.70	12.0	66.6	55.3	2.71 [2.20, 3.21]	3.37
S&P 500	3 017	1.11	−0.64	15.8	17.6	17.6	2.68 [2.08, 3.29]	2.80

- sd: the daily standard deviation (%); vol.: the annual volatility (%), with the actual A and with 252; $\hat{\alpha}$: the Hill tail index, $k = 2.5\%$ of n
- Interpretation: no: the left-tail indices are almost equal and their intervals overlap; Bitcoin is about four times more volatile, but its tail has the same shape: the difference is in scale

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B2: Ethereum and Solana [Proposed]

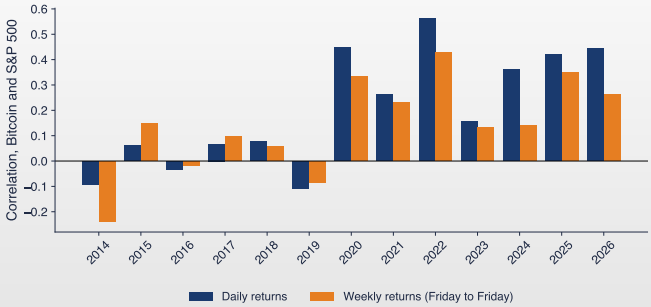
- **Question**: do the conclusions of B1 hold for two other large crypto assets? Model: B1.
- **Inputs**: Ethereum since 2016, Solana since April 2020; daily log returns in %
- **Tasks**
  1. Compute the four moments of each series.
  2. Annualise the volatility with 365 and with 252 days.
  3. Compute both Hill indices with their 95% intervals.
  4. Interpretation: which of the five assets of B1 and B2 has the heaviest left tail?
- **Report**: one table and two sentences
- Notebook: section B2

B3: Bitcoin and the S&P 500 in Calm and Crisis [Solved]

- **Question**: did the correlation between Bitcoin and equities change after 2020, and what happened in the 2020 and 2022 crises?
- **Inputs**: Bitcoin and S&P 500 closes since September 2014, joined on common days; daily and weekly (Friday to Friday) log returns
- **Tasks**
  1. Compute the correlation before 2020 and from 2020 on, and test their equality with the Fisher z test.
  2. Compute the correlation of daily and of weekly returns year by year and draw both.
  3. Compute the cumulative returns of both series in the COVID-19, Terra/Luna and FTX episodes.
  4. Compute Bitcoin's mean return on the worst 5% of S&P 500 days.
  5. Interpretation: does Bitcoin diversify an equity portfolio when it is most needed?
- **Report**: six numbers, the chart and two sentences
- Notebook: section B3

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B3: Solution (1/2) [Solved]



- Before 2020: $\hat{\rho} = 0.02$ ($n = 1\,330$); from 2020: $\hat{\rho} = 0.39$, 95% CI $[0.35, 0.43]$ ($n = 1\,687$); Fisher $z = 10.71$, $p < 0.001$

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B3: Solution (2/2) [Solved]

- Cumulative log returns (Bitcoin; S&P 500): COVID-19 -40.6%; -41.4%; Terra/Luna -23.0%; -5.4%; FTX -29.9%; 4.6%
- On the 151 worst S&P 500 days (mean -2.70%), Bitcoin returned on average -2.77%; weekly correlation over the whole period 0.19, daily 0.24
- Interpretation: no: since 2020 the correlation is significantly positive, and on the worst equity days Bitcoin falls as much as the S&P 500; crypto-specific crises (Terra, FTX) did not spread to equities

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B4: Bitcoin and Gold, Ethereum and the S&P 500 [Proposed]

- **Question**: is Bitcoin linked with gold, and does Ethereum behave like Bitcoin towards equities? Model: B3.
- **Inputs**: Bitcoin and gold (XAU/USD) since 2014; Ethereum and S&P 500 since 2016; prices joined on common days
- **Tasks**
  1. Compute both correlations before and after 2020 and test their equality.
  2. Compute the correlations of 2022 and 2025, daily and weekly.
  3. Compute the mean return of Bitcoin on the worst 5% of gold days, and of Ethereum on the worst 5% of S&P 500 days.
  4. Interpretation: is Bitcoin "digital gold"?
- **Report**: one table and two sentences
- Notebook: section B4

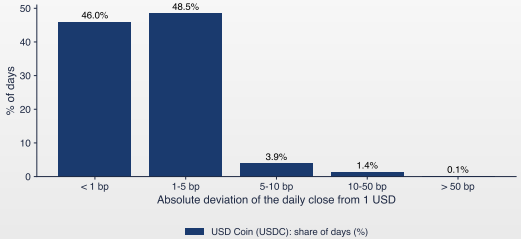
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B5: How Stable Is USDC? [Solved]

- **Question**: how far and for how long does USDC move away from 1 USD?
- **Inputs**: USDC daily closes and lows since 1 January 2021
- **Tasks**
  1. Compute the peg deviations in bp and their standard deviation and median absolute value.
  2. Compute the share of days in the classes below 1, 1–5, 5–10, 10–50 and above 50 bp, and draw them.
  3. Fit an AR(1) to the deviation and compute the half-life of a deviation.
  4. Compare the standard deviation before 2023 and after April 2023.
  5. Interpretation: can a treasurer treat USDC as cash?
- **Report**: six numbers, the chart and two sentences
- Notebook: section B5

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B5: Solution [Solved]



- $n = 2\,087$ days; sd 7.5 bp, median $|d_t|$ 1.1 bp; 1.5% of days beyond 10 bp, 0.14% beyond 50 bp; worst close -285 bp (11 March 2023), worst low -1226 bp
- AR(1): $\hat{\phi} = 0.290$ (se 0.021), half-life 0.56 days; sd 5.8 bp in 2021–2022, 1.5 bp after April 2023
- Interpretation: almost always: on a typical day USDC is within a few bp and deviations close within a day; but the tail is real: on one weekend of 2023 a holder could lose 3% at the close and 12% at the low

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B6: USDT and DAI [Proposed]

- **Question**: are USDT and DAI as stable as USDC? Model: B5.
- **Inputs**: USDT and DAI daily closes and lows since 1 January 2021
- **Tasks**
  1. Compute the standard deviation, the median absolute deviation and the shares beyond 10 and 50 bp.
  2. Find the worst close and the worst low of each coin, with their dates.
  3. Fit the AR(1) and compute the half-lives.
  4. Interpretation: which of the three coins keeps the peg best in normal times, and which in a crisis?
- **Report**: one table and two sentences
- Notebook: section B6

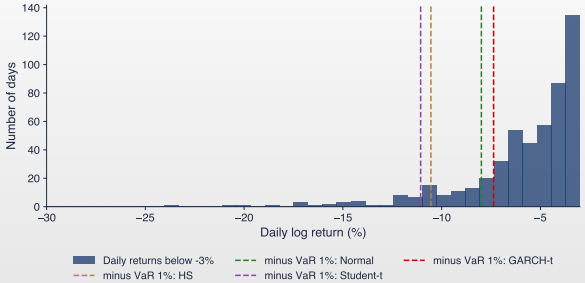
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B7: Risk of a Bitcoin Position [Solved]

- **Question**: how much can a position of 100 000 USD in Bitcoin lose in one day, and which model would have passed a backtest?
- **Inputs**: Bitcoin closes since September 2014, daily log returns in %
- **Tasks**
  1. Compute VaR 1% and ES 2.5% by historical simulation, Normal, Student-t (maximum likelihood) and GARCH(1,1)-t for the next day.
  2. Convert them into USD with $W(1 - e^{-v/\mathbf{100}})$.
  3. Backtest the one-day VaR 1% since 2018: historical simulation on the last 365 days and GARCH(1,1)-t re-estimated every 365 days; count the exceptions and run the Kupiec test.
  4. Interpretation: which number would you report to a risk committee?
- **Report**: a table of eight numbers, the chart, two test results and two sentences
- Notebook: section B7

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B7: Solution (1/2) [Solved]



- $n = 4\,384$; Student-t $\hat{\nu} = 2.23$; GARCH-t $\hat{\sigma}_{t+\mathbf{1}} = 2.83\%$, $\hat{\nu} = 3.18$; backtest since 1 January 2018 ($n = 3\,183$, expected 31.8): HS 29 exceptions, LR $= 0.26$ (p $= 0.61$); GARCH-t 41, LR $= 2.45$ (p $= 0.12$)

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B7: Solution (2/2) [Solved]

	HS	Normal	Student-t	GARCH-t
VaR 1% (%)	10.54	7.99	11.06	7.37
VaR 1% (USD)	10 006	7 679	10 474	7 105
ES 2.5% (%)	10.84	8.03	13.37	8.11

- Interpretation: report the historical VaR 1% (10 006 USD) as the long-run number and the GARCH-t VaR (7 105 USD) as today's number; the Normal VaR is 24% too low; both backtested models pass

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B8: Risk of an Ethereum Position [Proposed]

- **Question**: does the conclusion of B7 hold for Ethereum? Model: B7.
- **Inputs**: Ethereum closes since 2016; backtest since 1 January 2019
- **Tasks**
  1. Compute VaR 1% and ES 2.5% by the four methods of B7, in % and in USD.
  2. Backtest the two one-day VaR 1% forecasts and run the Kupiec test.
  3. Find the largest number of exceptions in one calendar year for each model.
  4. Interpretation: which model would you keep for Ethereum?
- **Report**: one table and two sentences
- Notebook: section B8

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C1: Did the Spot ETFs Change Bitcoin? [Proposed]

- **Question**: since 11 January 2024 US investors can hold Bitcoin through spot ETFs traded on weekdays; did Bitcoin's volatility, its weekend pattern or its link with equities change?
- **Inputs**: Bitcoin and S&P 500; two years before and two years after the launch; models: B1, B3
- **Tasks**
  1. Compute the annualised volatility in both windows and test the equality of variances (Brown–Forsythe).
  2. Compute the ratio of weekend to weekday variance in both windows.
  3. Compute the correlation with the S&P 500 and the GARCH(1,1)-t persistence in both windows.
  4. Interpretation: can a before–after comparison prove an effect of the ETFs?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the risks of Bitcoin and stablecoins. The answer:
- (a) Bitcoin's annual volatility is 55.3%: the daily standard deviation times sqrt(252).
- (b) The VaR 99% of 100,000 USD in Bitcoin is −10 543 USD.
- (c) On 11 March 2023 USDC lost 2.85 basis points.
- (d) TerraUSD was a fiat-backed stablecoin whose reserves were US Treasury bills.
- (e) Bitcoin is a safe haven: its correlation with the S&P 500 is 0.02.
- (f) Bitcoin returns are less volatile at weekends than on weekdays.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

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Points to Remember

- Annualise crypto with 365 days; never mix 252 and 365 in one ratio
- A fall of d needs a gain of $d/(1-d)$; crypto drawdowns of 75–85% need gains of 300–570%
- Peg deviations are measured in basis points; they are tiny in normal times and large in a run
- Since 2020 Bitcoin moves with equities; it is not a safe haven in this sample
- An AI answer is a draft: check the annualisation, the VaR level and sign, and the units
- Next
  - Lecture 14 develops these topics and adds blockchain basics, efficiency, bubbles and regulation
  - C1 can grow into a team project: a control group, intraday data, ETF flows
  - Reading: Franke, Härdle & Hafner (2019), Ch. 23; Liu & Tsyvinski (2021); Lyons & Viswanath-Natraj (2023)

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